# SmartUA

## Google Ads API Integration — Design Document

*Submitted in support of a Basic Access application to the Google Ads API*

Version 1.0 | Confidential — Internal Use Only

# 1. Tool Overview

SmartUA is an internal user-acquisition (UA) campaign management and automation platform built for our own growth team. It connects to the Google Ads API to read campaign performance data and to apply a small, well-defined set of pre-approved optimization actions. The tool is used exclusively by internal UA operators and is not exposed to external clients or the public.
This document describes the tool's purpose, intended audience, the scope of Google Ads API operations it performs, its architecture and data flow, and the security, approval, and rate-limiting controls that govern its use.

# 2. Intended Use and Audience

- Primary audience: internal UA / growth-team employees only.
- The application and its OAuth client are configured for internal use; no external user accounts are provisioned or supported.
- Use case: routine monitoring and optimization of our own Google Ads campaigns to improve return on ad spend (ROAS) and reduce manual effort.
- No data is resold, shared with third parties, or used for any purpose outside internal campaign management.

# 3. Functional Capabilities

## 3.1 Read (monitoring & reporting)

- Pull campaign, ad-group, and ad-level performance metrics: impressions, clicks, cost, installs, conversions, cost-per-install (CPI), and return on ad spend.
- Build internal dashboards and generate performance summaries for human reviewers.

## 3.2 Write (limited, pre-approved optimization)

- Pause or enable campaigns based on performance rules.
- Adjust a campaign's daily budget within configured bounds.
- Update an ad group's bid (e.g., CPC bid).
- Rotate ad creatives by pausing under-performing ads.

Every write action is derived from an explicit operator intent, validated by the platform, and gated by a human approval step (see Section 7) before it reaches the live API.

# 4. Google Ads API Usage

## 4.1 Authentication

- OAuth 2.0 with offline access; a long-lived refresh token is used for server-to-server calls (no interactive login at runtime).
- Granted scope: https://www.googleapis.com/auth/adwords (Google Ads API).
- Target customer (operating) account ID: 2046956901 (configured invisibly; not a user-facing field).

## 4.2 Read operations (GoogleAdsService.Search, GAQL)

- SELECT campaign.id, campaign.name, campaign.status, campaign_budget.amount_micros, metrics.* FROM campaign WHERE segments.date BETWEEN ...

- Analogous queries for ad_group and ad_group_ad to surface installs, conversions, and cost.

### 4.3 Write operations (GoogleAdsService.Mutate)

| Operation | API mutate performed |
| --- | --- |
| Update campaign status | CampaignOperation: status = ENABLED \| PAUSED |
| Update campaign budget | CampaignBudgetOperation: amount_micros = daily_budget * 1,000,000 |
| Update ad-set / ad-group bid | AdGroupOperation: cpc_bid_micros = bid * 1,000,000 |
| Rotate creative | AdGroupAdOperation: status = PAUSED on the target ad |

## 5. Architecture and Data Flow

1. An operator issues a natural-language intent through the SmartUA console (e.g., "pause the under-performing campaign").
2. The intent engine parses the request into a structured action.
3. The agent runtime plans concrete steps and estimates their expected impact.
4. Write actions are held at a human-in-the-loop approval gate (L1/L2) before any API call.
5. The Google Ads connector translates the approved action into the appropriate GAQL Search or Mutate call.
6. The API response is normalized, recorded, and fed back into the platform's analytics and memory for continuous learning.

The connector layer is isolated behind a common interface, with an automatic mock fallback that is used in development and whenever live credentials are absent — ensuring no accidental calls to production during testing.

## 6. Data Handling and Security

- Credentials: the OAuth 2.0 refresh token and client secret are stored server-side in an environment configuration file that is excluded from version control (.env, gitignored) and/or in an encrypted credentials table; they are never embedded in client code, logs, or error messages.
- No end-user Google account passwords are ever collected or stored by the tool.
- Token scope is limited to the adwords scope; the application requests only the access necessary for its function.
- Data accessed is limited to campaign performance aggregates (spend, installs, conversions). The tool does not process end-user personal data.
- Access to the tool itself is restricted to authenticated internal users.

## 7. Permissions and Human-in-the-Loop Controls

- All write operations require an explicit human approval before execution: low-risk changes (L1) and higher-risk changes (L2) follow a tiered approval policy.
- Every approved or rejected action is recorded in an audit log with the actor and timestamp.

- The default operating platform is configurable; the system safely falls back to a non-destructive mock mode when live credentials are not present.

## 8. Rate Limiting and Error Handling

- The connector respects Google Ads API rate limits and applies exponential backoff on 429 / retryable errors.
- Errors are captured and surfaced to operators; failed writes never leave the system in a partially corrupted state.
- A mock connector provides deterministic behavior for development and testing without consuming live API quota.

## 9. Compliance Commitment

- SmartUA is built to comply with the Google Ads API Terms of Service and the associated policy for API Clients.
- The tool is for internal use only; no data is resold, transferred to third parties, or used beyond the stated purpose.
- We commit to keeping the OAuth consent screen accurate and to maintaining security practices for stored credentials.

*— End of document —*